Sumaiya Dharsi

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EDUCATION

Northeastern University, College of Engineering, Boston, MA *Expected May 2028*

Bachelor of Science in Mechanical Engineering and Design

Related Courses: Cornerstone of Engineering, Statics, Thermodynamics, Mechanics of Materials, Dynamics, Designing Interactive Experiences

PROFESSIONAL EXPERIENCE

10Beauty **Burlington, MA**

Mechanical Engineering Co-op *January 2026 - June 2026*

- Prototyped components for a robotic manicure machine using SolidWorks, collaborating with the engineering team to enhance consistency and reliability across design iterations.
- Performed force and viscosity testing on material samples, analyzing results to drive iterative design refinements for the next build.
- Presented testing results and design recommendations to the engineering team to guide prototype development priorities.

Digitech Printers **East Meadow, NY**

Design Intern *May 2023 - August 2023*

- Produced custom apparel designs for 7+ clients using heat press and vinyl application techniques.
- Operated vinyl cutting equipment and heat press machinery, adjusting settings for different fabric types and designs.
- Created custom designs using Adobe Illustrator and Photoshop, gaining hands-on exposure to design workflow and production processes in a professional print environment.

PROJECTS

MatchaBot

Personal Project *January 2026 - Present*

- Engineered an automated matcha whisking device that replicates the traditional chasen's zigzag motion, performing gear calculations and mechanical design to convert rotational motion into whisking action.
- Integrated 3D-printed mechanisms with an Arduino microcontroller to control motion and achieve smooth, froth-free matcha.
- Designed a rack-and-pinion drive system with Arduino-controlled limit switches to precisely control the whisk's range of motion.

***Flexir*: Hand Physical Therapy Tool**

Cornerstone of Engineering *January 2025 - April 2025*

- Designed and prototyped a wearable rehabilitation device in SolidWorks with 3D-printed components for comfort and durability.
- Integrated Arduino flex sensors with MATLAB to collect and analyze real-time hand-mobility data.
- Conducted and analyzed 10 user tests to calibrate sensors, validate accuracy, and demonstrate measurable progress tracking for at-home therapy use.

Wreck-It Ralph: Building Disaster

Cornerstone of Engineering *September 2024 - December 2024*

- Developed an interactive educational game using C++, Arduino, and laser-cut components to introduce children to structural engineering concepts.
- Conducted user experience testing, gathering feedback to refine gameplay and improve learning outcomes.
- Gained practical prototyping and teamwork experience through collaborative, hands-on design.

SKILLS

Programming Languages: C++, HTML, CSS, MATLAB
CAD & Mechanics: SolidWorks, Arduino, 3D printing (SLA/FDM), Laser Cutting, Soldering
Digital Tools: Microsoft Suite, Adobe Suite
Spoken Languages: English (Fluent), Chinese (Proficient), Gujarati (Proficient)
Interests: Painting (Acrylic Medium), Pottery, Volleyball, Soccer